Experiment – 8

Workflow

Workflow Automation using Zoho Creator

Department of Computer Science and Engineering
Student Name: Sachin Sarvanan
Roll No: 240701449
Application Name: Purchase Request / Procurement Application

Objective

To automate key tasks in Zoho Creator by setting up form workflows triggered by record creation or editing. A Deluge script checks whether a Purchase Request fits within the available budget, while an email action can notify relevant users.

Introduction

A Zoho Creator workflow runs configured actions automatically when a specified event takes place. This helps handle routine checks, calculations, record changes, and notifications with less manual effort. A form workflow is set up by choosing its form, the record event (Created, Edited, Created or Edited, or Deleted), and the form event, such as form load, field input, or successful submission. After the trigger, it can execute Deluge code, send an email, update or add records, or display a message.

Workflow Components in Use

Component	Description	Used in this Experiment
Form	The form where the workflow is defined	Purchase Request
Record Event	The record change that triggers the workflow	Created or Edited
Form Event	The stage of the form when actions run	Form load, field input, successful form submission
Action – Deluge script	Custom logic written in Deluge	budget-check script updates Budget Status
Action – Send email	Sends a notification from the application	Notification when a record is added or updated

Table 8.1: Workflow components used in the application.

Purchase Request Form: Workflow Setup

The Purchase Request form has five enabled workflows, with one action configured in each workflow.

Workflow Name	Record Event	Form Event	Actions
Decision Field Access	Created or Edited	Load of the form	1 Action
calculate	Created or Edited	User input of Items → Quantity	1 Action
Finance Approval Status Update	Created or Edited	Successful form submission	1 Action
budget-check	Created or Edited	Successful form submission	1 Action
Hod_Approval	Created or Edited	Successful form submission	1 Action

Table 8.2: Workflows on the Purchase Request form.

Zoho Creator Screenshots

The pages below show the original Zoho Creator screenshots for the workflow configuration.

Screenshot 1 – Workflow List

The Workflow area groups automations according to the form and the event that triggers them. Purchase Request is listed among forms with workflows; Approval History, Budget, Department, Purchase Request Items, and Vendor are listed as forms without workflows. Navigation tabs also provide access to Schedules, Approvals, Payments, Blueprints, Batch workflows, and Functions.

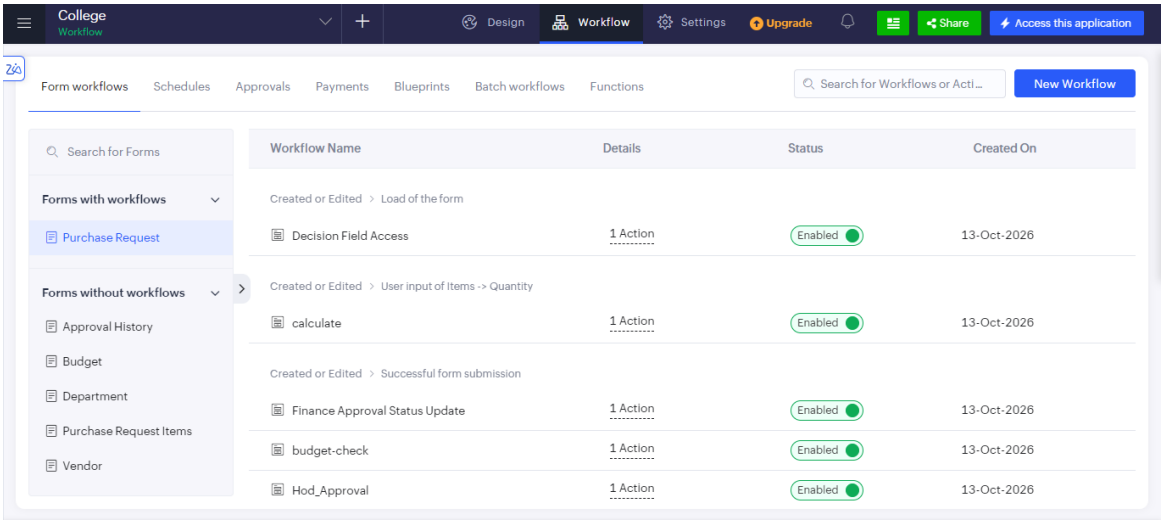


Figure 1: Form workflows of the Purchase Request form, all enabled.

Screenshot 2 – Creating a Workflow

To add a workflow, choose New Workflow, select the target form and record event, specify the form event that should trigger it, enter a workflow name, and then select Create Workflow.

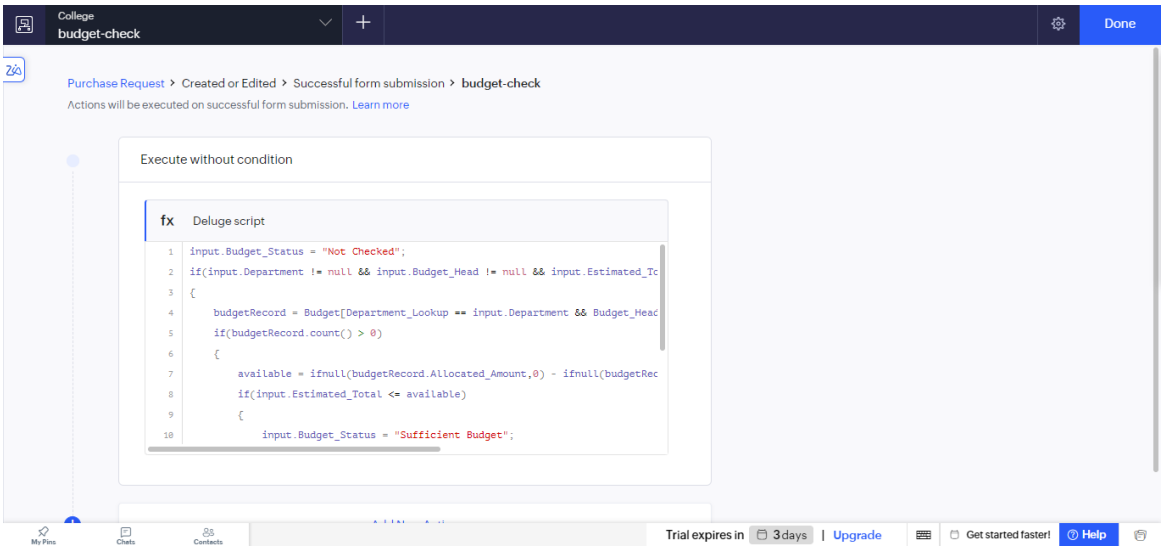


Figure 2: Workflow creation settings.

Screenshot 3 – Workflow Action

The budget-check workflow is linked to Purchase Request → Created or Edited → Successful form submission. Its Execute without condition action runs the Deluge script after the form is submitted successfully.

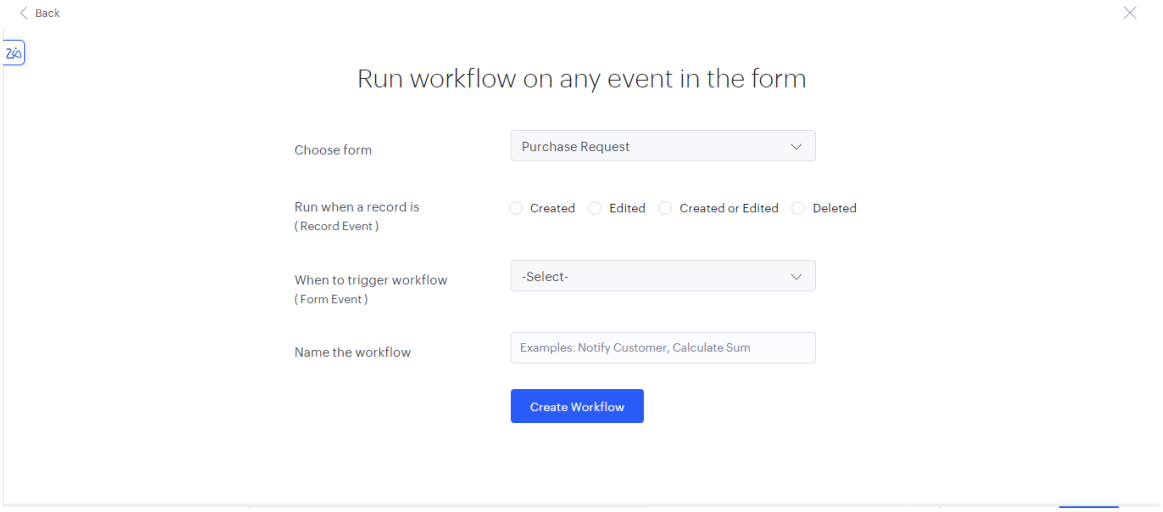


Figure 3: budget-check workflow with a Deluge script action.

Screenshot 4 – Budget-Check Deluge Script

The Deluge editor includes a side panel with options such as Condition and Data Access. The script compares the request's estimated total with the department's remaining budget and then updates the Budget Status field.

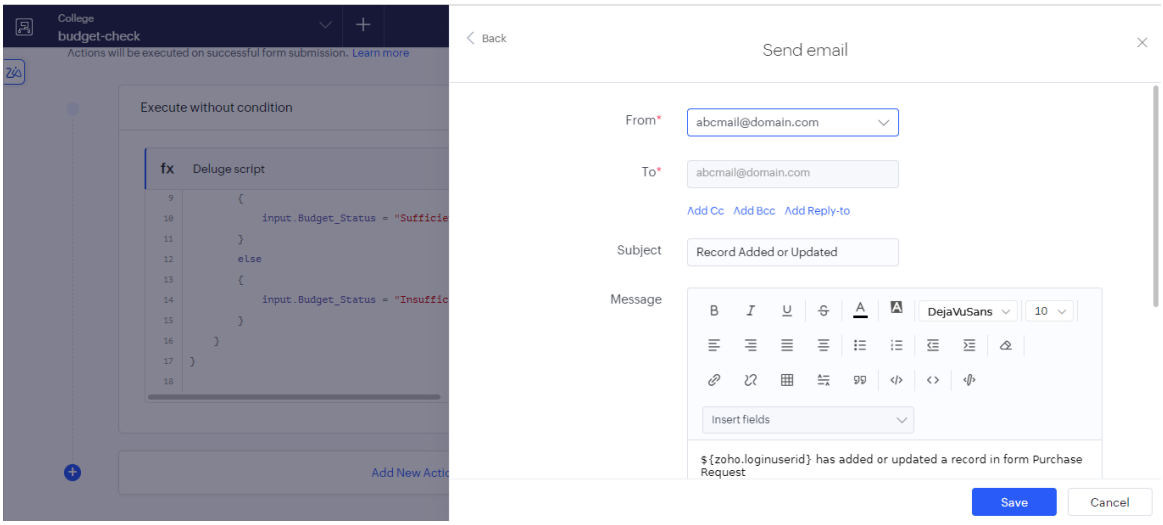


Figure 4: Deluge script editor showing the budget-check script.

Screenshot 5 – Send Email Action

A Send email action can also be added through Add New Action. Configure the sender, recipient, subject, and message, and use Insert fields to include form values or system variables. In the example, zoho.loginuserid identifies the user who added or updated the Purchase Request record.



Figure 5: Send email action with From, To, Subject, and Message.

Budget-Check Script

```
input.Budget_Status = "Not Checked";
if(input.Department != null && input.Budget_Head != null && input.Estimated_Total != null)
{
budgetRecord = Budget[Department_Lookup == input.Department
&& Budget_Head == input.Budget_Head];
if(budgetRecord.count() > 0)
{
available = ifnull(budgetRecord.Allocated_Amount,0)
- ifnull(budgetRecord.Reserved_Amount,0)
- ifnull(budgetRecord.Spent_Amount,0);
if(input.Estimated_Total <= available)
{
input.Budget_Status = "Sufficient Budget";
}
else
{
input.Budget_Status = "Insufficient Budget";
}
}
}
```

Script Explanation

1. The Budget Status field is first reset to Not Checked.

2. The script continues only when Department, Budget Head, and Estimated Total have values.

3. It searches the Budget form for a record matching the selected department and budget head.

4. Available funds equal the allocated amount minus reserved and spent amounts; ifnull() treats missing amounts as zero.

5. When the estimated total is within the available amount, the status becomes Sufficient Budget; otherwise, it is set to Insufficient Budget.

Observations

The workflow automates several steps in the Purchase Request process. After submission, the budget-check script retrieves the matching department budget and marks the request as Sufficient or Insufficient, reducing the need for manual comparisons. Other workflows control field access on form load, recalculate values when item quantity changes, and update approval status after submission. The email action demonstrates how notifications can be configured alongside other workflow actions. Choosing the appropriate record and form events ensures that each action runs at the required stage.

Conclusion

Five workflows were configured for the Purchase Request form in Zoho Creator. The budget-check workflow runs a Deluge script after successful submission to determine budget sufficiency, and the Send email action is configured to notify users when a record is added or updated.